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Essential, Non-Essential, and Toxic Elements in the Muscle of Meagre (*Argyrosomus regius*) from the Tagus Estuary (Portugal)

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Abstract

Monitoring trace metals in commercially important fish species provides an early warning of anthropogenic contamination and potential risk to consumers. This study semi-quantified and quantified essential, non-essential, and toxic elements in the muscle of wild meagre (*Argyrosomus regius*) captured in the Tagus estuary (Portugal), which is used as a nursery and spawning aggregation area. Dry muscle was microwave-digested and analyzed using inductively coupled plasma–optical emission spectroscopy. Semi-quantified screening detected Al, B, Ca, Fe, K, Mg, Na, P, S, Si, Sr, and Ti, and eight elements were determined using multielement calibration (As, Cr, Cu, Hg, Mn, Ni, Se, and Zn); Cd, Pb (toxic elements), Co, and Mo were not detected in this study. Arsenic was detected in all individuals, with a minimum value of 0.348 mg/kg wet weight. A mercury level above the European Commission regulatory limit (0.5 mg/kg wet weight) was only detected in one individual, corresponding to 2% of the samples. Although other metals remain well below regulatory limits, continued biomonitoring is recommended to track temporal trends and safeguard seafood safety in transitional coastal systems, which is important for commercially relevant fish species.

Keywords: Sciaenidae; heavy metals; inductively coupled plasma–optical emission spectroscopy (ICP-OES); food contamination; food safety; one health

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1. Introduction

The detection of heavy metals in aquatic systems is well-established, especially in transitional water bodies and freshwater, usually due to anthropogenic sources. Anthropogenic inputs of heavy metals are predominantly associated with urban and industrial discharge, maritime traffic, and recreational usage of water bodies, as well as food production sectors

(e.g., fisheries, aquaculture, agriculture, and industrial food processing) [1–4]. These elements exhibit physicochemical properties, such as persistence, chemical inertness, and non-degradability, which facilitate their long-term environmental presence [5,6].

In aquatic ecosystems, heavy metals tend to accumulate in sediments due to their chemical properties [7,8]. However, several studies also report their presence in the water column [9]. In fish, the gills, skin, and digestive tract are the main entry points for heavy metals into the body. In recent years, several authors acknowledged the presence of these elements in seafood and assessed, in these matrices, the risk to consumers [10–12].

The estuarine ecosystems are usually exposed to a diverse array of anthropogenic stressors, including agricultural runoff, industrial effluents, and domestic wastewater discharge, leading to significant alterations in estuarine water quality and subsequent ecosystem impacts [13].

The Tagus Estuary, situated along the western coastline of Portugal, spans an area of approximately 320 km^2, and it is recognized as one of Europe's largest estuarine systems [14]. Its shores encompass industrial, agricultural, and densely urbanized zones [15]. Notably, the Tagus estuary still functions as a critical nursery habitat for multiple economically significant fish species [16,17].

Argyrosomus regius (Asso, 1801), commonly known as the meagre, is a demersal predatory fish from the Sciaenidae family and ranks among the largest teleost fishes inhabiting European marine waters [18]. This migratory species initiates its life cycle in estuarine habitats, predominantly during spring and summer, followed by offshore migration in autumn for overwintering [19]. The spawning migration to estuarine or coastal areas occurs between mid-April and the end of May [20]. In the specific case of the Tagus estuary, adult meagre enter the estuary in February/March, reproduce in April/June, and stay in the estuary until August/September [21]. As a food product, it presents a high number of qualities, such as a large market size, a low fat content [22–24], high-quality lipids [22–24], high protein levels, and a pleasant flavor [23]. Nevertheless, all these assumptions were made when analyzing meagre from aquaculture and not wild individuals.

In Portugal, *A. regius* is a commercially valuable species, with landings in 2024 totaling 323 metric tons, of which 76% were sourced from the Tagus estuary and adjacent coastal areas, with an average first sale price of 9.6 EUR/kg on average (*DGRM unpublished data*).

The analysis of the elemental composition in the muscle tissue of meagre has considerable scientific relevance, as this species represents a food product with high nutritional value [25], containing nutritionally important elements such as potassium (K), magnesium (Mg), and other biologically significant trace minerals. However, depending on the environment, it can also act as a vector for exposure to heavy metals (e.g., mercury and arsenic), as its food chain includes various pelagic, demersal, and cephalopod species, among others, which poses a risk of contamination, namely through the accumulation of heavy metals, which could cause potential health risks for consumers. Indeed, Chiesa et al. [26] and Maulvault et al. [27] found levels of heavy metals and other potentially harmful metals in mussels, clams, and mullets (*Liza aurata*) from the Tagus estuary area.

Information regarding the element composition of meagre flesh from wild stocks in the Tagus estuary is scarce. To our knowledge, only a previous study conducted by Chaguri et al. [28] investigated the levels of different elements, such as mercury (Hg), lead (Pb), zinc (Zn), and iron (Fe), in meagre sourced from both aquaculture and wild stocks. However, the focus was limited to small meagre under 1 kg, as well as those between 2–6 kg in weight, thereby excluding adults exceeding 8 kg.

This study aims to quantify the concentrations of essential, non-essential, and toxic trace elements in wild meagre captured inside the Tagus estuary from the juvenile to adult

stages, and to assess the potential health risks posed to consumers through the consumption of this species.

2. Materials and Methods

2.1. Collecting, Sampling, and Lyophilizing the Muscle of Meagre Samples from the Tagus Estuary

All 55 wild meagre individuals were captured by local fishermen in the Tagus estuary between June and September 2020, with the majority captured in June/July. The sample consisted of individuals from both genders (32 males and 23 females), divided into three-dimensional classes or ontogenic stages (less than 2 kg—small, 2 to 8 kg—large juvenile, and more than 8 kg—adults). For easier understanding, in this work, small juvenile, large juvenile, and adult individuals are named as small, juvenile, and adult, respectively. Captured specimens were quickly transported (less than a 30 min trip, 16 km distance) to the laboratory (Marine and Environmental Sciences Centre, Faculty of Sciences of the University of Lisboa), where the fish were filleted, and genders were identified through macroscopical observation of the gonads. The skinless fillets (i.e., all individual portions of approximately 10 g from muscles in the proximity of the mid-dorsal line, in the left flank of the fish, close to the dorsal fin) were collected, washed with physiological saline, homogenized individually, and weighed. Subsequently, a portion was deep-frozen in liquid nitrogen and immediately stored at −80 °C until further laboratory processing. Frozen samples were lyophilized in an Edwards freeze-dryer for 48 h and subsequently weighed.

2.2. Reagents

All the reagents used were of analytical grade, as purchased. The standards used phosphorus (P) and tungsten (W) as certified material references for the inductively coupled plasma–optical emission spectroscopy (ICP-OES) analysis solution (CPAChem, Ivanka Terzieva Str. 6065, Bogomilovo Bulgaria, 1 g/L); aluminum (Al), antimony (Sb), bismuth (Bi), cadmium (Cd), chromium (Cr), cobalt (Co), lithium (Li), manganese (Mn), nickel (Ni), potassium (K), selenium (Se), silicon (Si), strontium (Sr), thallium (Tl), tin (Sn), yttrium (Y), and zinc (Zn) were used as certified material references for the atomic absorption spectroscopy (AAS) analysis solution (CPAChem, Ivanka Terzieva Str. 6065 Bogomilovo Bulgaria, 1 g/L); arsenic (As), lead (Pb), and mercury (Hg) were used for the standard solution (TraceCert®) for ICP analysis (Supelco, 595 North Harrison Road Bellefonte, PA 16823 USA, 1 g/L); zirconium (Zr) was used as the standard solution (Certipur®) for ICP analysis (Supelco, 595 North Harrison Road Bellefonte, PA 16823 USA, 1 g/L); and barium (Ba), boron (B), calcium (Ca), copper (Cu), iron (Fe), magnesium (Mg), molybdenum (Mo), silver (Ag), sodium (Na), sulfur (S), titanium (Ti), and vanadium (V) were used as the standard solution (TraceCert®) for AAS analysis (Supelco, 595 North Harrison Road Bellefonte, PA 16823, USA, 1 g/L). The digestion solution: HNO_3 Suprapur® for trace analyses (67–69%, Carlo Erba, Via R. Merendi, 22, 20010 Cornaredo (MI), Italy), and H_2O_2 $\geq$ 30%, for trace analysis (Supelco, 595 North Harrison Road Bellefonte, PA 16823, USA), were used as digestion solutions. Milli-Q water (sensitivity of 18.2 MΩ/cm) was obtained from Millipore Purification System (Watermax W1, Diwer Technologies, Merck Life Science S.L.U. Portugal, Algés, Portugal) and used for standard and sample dilutions. The most suitable standard reference material (ERM-BB422) with dried fish muscle, the species *Pollachius* virens (Saithe), was purchased from the National Institute of Standards and Technology (NIST, Gaithersburg, MD, USA).

2.3. Sample Preparation: Extraction with HNO_3:H_2O_2

To obtain different elements from meagre muscle, the lyophilized samples were wet-digested with a solution of nitric acid and hydrogen peroxide (3:1 v/v) in a microwave

digestion system (Multiwave Go Plus, Anton Paar Germany GmbH, Hellmuth-Hirth-Strasse 6, 73760 Ostfildern-Scharnhausen, Germany) equipped with a twelve-position rotor. The sample containing 0.200 ± 0.005 g of dry weight (dw) was weighed in a clean and decontaminated microwave Teflon digestion vessel (50 mL), and 8.0 mL of solution was subsequently added. The sealed vessel containers were placed in the rotor of a microwave oven in appropriate positions, and the samples were extracted according to the following mineralization program: a ramp temperature from room temperature to 120 °C for 10 min, followed by a 60 min hold at 120 °C, and then a decrease in temperature from 120 °C to 70 °C for 10 min, with a constant power supply of 300 W. After the end of the microwave digestion program, the sample solutions were cooled and transferred quantitatively into calibrated volumetric flasks (20 mL) and diluted with Milli-Q water. The flasks with the solutions were closed to preserve the real concentration of elements, and stored at 4 °C before the ICP-OES analysis.

2.4. Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) Analysis

The analysis was performed on a PlasmaQuant 9100 Elite ICP-OES (Analytik Jena, Konrad-Zuse-Strasse 1, 07745 Jena, Germany). The sample introduction components, as well as the instrumental settings, were selected to achieve a high level of sensitivity for trace elements in the expected high-salt-containing matrices. In conjunction with this instrument, a Teledyne CETAC ASX-560 autosampler (Teledyne Labs, 14306 Industrial Rd, Omaha, NE 68144, USA) was used. The analysis conditions were: plasma power of 1200 W, plasma gas flow of 12 L/min, auxiliary gas flow of 0.5 L/min, a conical concentric nebulizer, 1 mL/min in borosilicate, with a gas flow of 0.6 L/min, a cyclonic spray chamber with 50 mL in borosilicate, and outer and inner tubes in quartz, a quartz injector, ID: 2 mm, a sample tube in PVC, pump rate of 1 mL/min, fast pump at 4 mL/min, measuring delay of 45 s, rinse time 15 s, and a 0 mm torch position. For quantification, a standard solution of yttrium was added to each sample for a final concentration of 0.1 mg/L. The plasma in the ICP-OES system was made by partially ionizing argon gas, and the line and plasma view used for each element are described in Table 1.

Table 1. Parameters used for individual element analysis (line and plasma view), limit of detection (LOD), quantification (LOQ) in µg/L, and type of quantification (semi-quantification/quantification, described below) used in ICP-OES analysis. SQ and Q denote semi-quantification and quantification, respectively.

Element	Line (nm)	Plasma View	LOD	LOQ	Type	Element	Line (nm)	Plasma View	LOD	LOQ	Type
Ag	328.06	axial	12.17	-	SQ	Na	589.592	Radial	18.54	-	SQ
Al	396.152	Axial	2.69	-	SQ	Ni	231.604	Axial	2.49	7.53	Q
As	188.979	Axial	2.89	8.77	Q	P	213.618	Axial	406.68	-	SQ
B	249.773	Axial	2.26	-	SQ	Pb	220.353	Axial	3.89	11.78	Q
Ba	233.527	axial	12.92	-	SQ	S	182.565	Axial	24.01	-	SQ
Bi	223.061	Axial	7.77	-	SQ	Sb	217.581	Axial	18.14	-	SQ
Ca	315.887	Radial	36.20	-	SQ	Se	196.028	Axial	5.28	15.99	Q
Cd	228.802	Axial	2.01	6.08	Q	Si	251.611	Axial	1.21	-	SQ
Co	228.615	Axial	2.50	7.58	Q	Sn	189.927	Axial	24.87	-	SQ
Cr	205.552	Axial	1.43	4.34	Q	Sr	407.771	Axial	19.04	-	SQ
Cu	213.598	Axial	3.59	10.88	Q	Ti	336.122	Axial	0.84	-	SQ
Fe	238.204	Axial	0.43	-	SQ	Tl	190.796	Axial	18.19	-	SQ
Hg	184.886	Axial	6.49	19.67	Q	V	311.838	Axial	3.44	-	SQ
K	766.491	Radial	281.51	-	SQ	W	239.709	Axial	4.56	-	SQ
Li	670.791	Radial	5.37	-	SQ	Y	371.030	Axial	-	-	IS
Mg	285.213	Radial	123.21	-	SQ	Zn	206.200	Axial	1.67	5.05	Q
Mn	257.610	Axial	3.42	10.37	Q	Zr	343.823	Axial	17.65	-	SQ
Mo	202.030	Axial	1.39	4.21	Q						

Semi-quantification (SQ); Quantification (Q); Internal Standard (IS).

2.5. Method Calibration and Validation

The method of standard addition calibration was run for the semi-quantification of 34 elements, and 12 of these were quantified. For semi-quantification, two spikes of As, Ag, Al, B, Ba, Bi, Cd, Co, Cr, Cu, Fe, Hg, Li, Mn, Mo, Ni, Pb, Sb, Se, Si, Sn, Sr, Ti, Tl, V, W, Zn, and Zr were added at levels of 0.0, 0.1, and 1.0 mg/L into a 0.200 dw of mixture of real meagre muscle. The same methodology was performed for other elements in different concentrations, such as K and P, where 0.0, 10.0, and 100.0 mg/L were used; for Ca and Mg, 0.0, 1.0, and 10.0 mg/L were used; for Na, 0.0, 2.0, and 20.0 mg/L were used; and for S, 0.0, 0.2, and 2.0 mg/L were used. A three-point standard addition calibration curve was prepared for the analysis of each dataset. For quantification, eight spikes of As, Cd, Co, Cr, Cu, Hg, Mn, Mo, Ni, Pb, Se, and Zn were added at levels of 0.00, 0.002, 0.005, 0.01, 0.02, 0.04, 0.06, 0.08, and 0.1 mg/L into a 0.200 dw mixture of real meagre muscle. A nine-point standard addition calibration curve was prepared for the analysis of each dataset. For semi-quantification, each element was measured once, for 1 s, and for quantification, each element was measured 3 times for 10 s, and the internal standard for 3 s. The LOD and the LOQ for the ICP-OES instrument and method were calculated after three consecutive runs by multiplying the standard deviation obtained. The instrumental LOD and LOQ were run with blank solutions, injected three times. To validate the microwave-assisted digestion method followed by the ICP-OES analysis for the quantification of elements, the standard reference material (ERM®-BB422), dried powdered fish muscle (*Pollachius virens* (Saithe), was used. Element determination was carried out after wet digestion in a closed-vessel microwave system and ICP analysis. The ERM®-BB422 was used to validate the microwave-assisted digestion followed by the ICP method for all element quantification tests. The NIST certified values for several elements are based on dw. The analytical procedure performed with ERM®-BB422 was the same as for the sample analysis.

2.6. Target Hazard Quotient (THQ), Hazard Index (HI), and Target Carcinogenic Risk (TR) Calculations

For a more uniform comparison of the potential health risk due to the ingestion of contaminants, US/EPA created a Target Hazard Quotient (THQ) [29]:

$$THQ = \frac{C \times ED \times EP \times FIR}{RfD \times BW \times AT} \quad (1)$$

This index is used to perform calculations by considering the following parameters: the concentration of elements in food (C, mg/kg) and the food ingestion rate (FIR), which is the daily consumption of fish (kg/day). In Portugal, the estimated FIR in 2022 was 0.149 kg/day (54.33 kg/year) [30], and the body weight (BW, kg) usually used is 70 kg; ED denotes the years of exposure frequency (years), and in Portugal, for the period 2022–2024, the average value was 81.49 years [31]; EP is the exposure frequency, equal to 365 days/year; the oral reference doses (RfD) refers to mg/kg day [32]; and average time (AT) refers to the average exposure time for non-carcinogens (AT = ED × EP).

If THQ $\geq$ 1, it indicates potential health hazards associated with the consumption of certain foods, and it is calculated for each element. The Hazard Index (HI) is a sum of all THQ values in one sample:

$$HI = THQa + THQb + \cdots + THQz \quad (2)$$

Exposure to carcinogens over a lifetime can enhance the probability of cancer, referred to as the Target Carcinogenic Risk (TR), and can be calculated using the following equation [33]:

$$TR = \frac{ED \times EP \times FIR \times C \times CSF}{BW \times AT} \quad (3)$$

If the values are between 10^{-6} and 10^{-4}, it indicates an acceptable cancer risk; if the values are below 10^{-6}, the risk is considered negligible. If the values are above 10^{-4}, it denotes a high target cancer risk.

3. Results

3.1. Semi-Quantification of Elements in the Muscle

In total, thirty-four elements were investigated in this study. Among these, twenty-two were just semi-quantified, and from those, nine were not detected in all samples, within the analytical limits, as presented in Table 2. The remaining twelve elements (the elements that were quantified) were also semi-quantified to estimate the range of concentration for the calibration curve for those elements; these results are not presented in this work. Detailed information can be found in Supplementary Table S1.

Table 2. Results of the ICP-OES analysis, expressed in mg/kg wet weight, for the semi-quantification of each element in meagre muscle, and the results of concentrations obtained by Chaguri et al. [28] for some of the elements analyzed in dry weight.

mg/kg Wet Weight	Range (Min.–Max.)	Small			Juvenile			Adult			Chaguri et al. [28] ☼
		Female	Male	Total	Female	Male	Total	Female	Male	Total	Dry Weight
Ag	-	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	-
Al	0.75–70.46	2.83 ± 1.01	4.38 ± 6.27	3.83 ± 5.12	10.38 ± 17.35	9.52 ± 12.40	9.89 ± 14.74	20.19 ± 20.90	11.91 ± 12.70	15.64 ± 17.38	-
B	0.03–31.98	0.74 ± 0.81	1.43 ± 1.08	1.21 ± 1.05	0.90 ± 0.83	1.37 ± 2.16	1.17 ± 1.74	4.63 ± 9.73	2.54 ± 1.85	3.48 ± 6.75	-
Ba	-	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	-
Bi	-	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	-
Ca	55.82–195.91	71.18 ± 4.94	74.89 ± 7.82	73.57 ± 5.12	82.97 ± 40.37	70.18 ± 5.87	75.66 ± 2754	70.57 ± 15.48	68.54 ± 6.38	69.45 ± 11.46	(0.51 ± 0.82) × 10^3
Fe	0.65–11.50	1.16 ± 0.36	1.75 ± 1.06	1.54 ± 0.92	2.87 ± 3.15	1.81 ± 0.95	2.27 ± 2.24	1.57 ± 0.67	1.56 ± 0.65	1.57 ± 0.66	11.34 ± 1.68
K	(2.43–3.82) × 10^3	(3.44 ± 0.14) × 10^3	(3.46 ± 0.20) × 10^3	(3.46 ± 0.18) × 10^3	(3.47 ± 0.17) × 10^3	(3.46 ± 0.20) × 10^3	(3.46 ± 0.19) × 10^3	(3.20 ± 0.35) × 10^3	(3.42 ± 0.12) × 10^3	(3.32 ± 0.27) × 10^3	(19.44 ± 0.82) × 10^3
Li	-	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	-
Mg	203.78–282.41	251.53 ± 7.93	259.37 ± 10.99	256.56 ± 10.69	253.38 ± 13.42	255.75 ± 12.15	254.73 ± 12.77	232.86 ± 13.76	240.96 ± 10.50	237.32 ± 12.73	-
Na	197.93–(1.11 × 10^3)	357.47 ± 130.59	379.75 ± 52.85	371.79 ± 89.34	355.53 ± 83.94	434.75 ± 123.95	400.80 ± 115.48	633.72 ± 263.45	516.10 ± 170.45	569.03 ± 225.03	-
P	(1.37–1.83) × 10^3	(1.61 ± 0.07) × 10^3	(1.64 ± 0.09) × 10^3	(1.63 ± 0.09) × 10^3	(1.61 ± 0.10) × 10^3	(1.66 ± 0.05) × 10^3	(1.64 ± 0.08) × 10^3	(1.50 ± 0.12) × 10^3	(1.53 ± 0.09) × 10^3	(1.51 ± 0.11) × 10^3	-
S	(1.40–2.16) × 10^3	(1.89 ± 0.16) × 10^3	(1.87 ± 0.15) × 10^3	(1.88 ± 0.15) × 10^3	(1.78 ± 0.17) × 10^3	(1.74 ± 0.18) × 10^3	(1.76 ± 0.18) × 10^3	(1.62 ± 0.14) × 10^3	(1.60 ± 0.09) × 10^3	(1.61 ± 0.12) × 10^3	(9.38 ± 1.57) × 10^3
Sb	-	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	-
Si	2.47–8.32	3.36 ± 0.23	3.70 ± 0.87	3.58 ± 0.73	3.33 ± 0.50	4.12 ± 1.51	3.78 ± 1.25	3.48 ± 0.85	3.35 ± 0.82	3.41 ± 0.83	-
Sn	-	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	-
Sr	N.D.–0.50	N.D.	N.D.	N.D.	0.50	N.D.	0.50	N.D.	N.D.	N.D.	-
Ti	N.D.–1.14	0.06 ± 0.04	0.05 ± 0.03	0.05 ± 0.03	0.05 ± 0.03	0.19 ± 0.31	0.13 ± 0.24	0.04 ± 0.01	0.10 ± 0.12	0.07 ± 0.09	-
Tl	-	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	-
V [Δ]	N.D.–0.11	N.D.	N.D.	N.D.	0.11	N.D.	0.11	N.D.	N.D.	N.D.	-
W	-	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	-
Zr	-	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	-

N.D. (Not Detected). ☼Values presented in dry matter for individuals from a range of 2 to 7 kg of total body weight, [Δ] detected only in 1 sample.

Titanium (Ti) and boron (B) were detected in the majority of the analyzed samples, with concentrations generally remaining below 100 µg/kg wet weight (ww) and 1 mg/kg *ww*, respectively. Iron (Fe), aluminum (Al), and silicon (Si) were also identified, although at low concentrations within the mg/kg ww range (Table 2). In contrast, several elements essential to human physiology, including calcium (Ca), magnesium (Mg), sodium (Na), phosphorus (P), sulfur (S), and potassium (K), were present in relatively high concentrations in the muscle tissue of meagre, which is expected because these elements are macro-elements in food (Table 2). Strontium (Sr) and vanadium (V), however, were detected in only a single sample, indicating their low prevalence in the studied population.

3.2. Quantification of Toxic and Other Elements in Meagre Muscle

Quantification was conducted for twelve elements, including the heavy metals arsenic (As), mercury (Hg), lead (Pb), and cadmium (Cd), as well as other potentially hazardous metals, whose toxicity is concentration-dependent, specifically cobalt (Co), chromium (Cr), copper (Cu), manganese (Mn), molybdenum (Mo), and nickel (Ni), in addition to selenium (Se) and zinc (Zn) (Table 3). Considering the presence of both port-related activities and upstream industrial discharges along the Tagus River, elevated concentrations of Zn, Ni, and Cu in the aquatic environment are likely [34]. More detailed information is present in Supplementary Table S2.

Table 3. Results of the ICP-OES analysis, expressed in mg/kg ww (range, means, and standard deviation), for the quantification of each element in meagre muscle, the reference material (measured, certified, and recovery), and the results of the concentrations obtained by Chaguri et al. [28] for some of the elements analyzed (dry weight).

mg/kg Wet Weight	Range (min.–max.)	Small			Juvenile			Adult			Reference Material			Chaguri et al. [28] Dry Weight
		Female	Male	Total	Female	Male	Total	Female	Male	Total	Measured	Recovery (%)	Certified	
As	0.35–2.54	1.18 ± 0.43	1.07 ± 0.38	1.11 ± 0.40	1.04 ± 0.62	0.79 ± 0.39	0.90 ± 0.52	0.940 ± 0.37	0.72 ± 0.27	0.82 ± 0.34	13.8 ± 0.2	108.9 ± 1.4	12.70 ± 0.70	9.61 ± 1.58
Cd	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	-	(7.50 ± 1.80) × 10^{-3}	0.02 ± 0.01
Co	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	-	-	-
Cr	<LOD–<LOQ	<LOD	<LOD	<LOD	<LOD	<LOQ	<LOQ	<LOD	<LOD	<LOD	<LOD	-	-	-
Cu	<LOD–0.53	<LOD	<LOD	<LOD	0.48	0.50 ± 0.02	0.50 ± 0.02	<LOD	<LOD	<LOD	<LOD	-	1.67 ± 0.16	2.68 ± 0.30
Hg	<LOD–0.66	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	0.57 ± 0.08	0.57 ± 0.08	<LOQ	-	(6.01 ± 3.00) × 10^{-1}	(1.66 ± 0.2) × 10^{-3}
Mn	<LOD–<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	<LOD	-	(36.80 ± 2.80) × 10^{-2}	-
Mo	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	-	-	-
Ni	<LOQ	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOQ	<LOQ	<LOD	-	-	-
Pb	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD	-	-	0.12 ± 0.05
Se	<LOQ–0.73	0.57 ± 0.09	0.50 ± 0.08	0.52 ± 0.09	0.56 ± 0.10	0.51 ± 0.10	0.54 ± 0.10	0.50 ± 0.09	0.59 ± 0.10	0.56 ± 0.11	<LOQ	-	1.33 ± 0.13	2.05 ± 0.21
Zn	2.23–8.65	3.01 ± 0.51	3.05 ± 0.32	3.03 ± 0.40	2.85 ± 0.36	3.09 ± 0.54	2.99 ± 0.48	3.07 ± 0.66	3.45 ± 1.71	3.28 ± 1.36	17.4 ± 0.1	108.7 ± 0.4	16.00 ± 1.10	14.53 ± 0.47

<LOD (Limit of Detection), <LOQ (Limit of Quantification).

For all elements, calibration curves were established, and both limits of detection (LOD) and limits of quantification (LOQ) were determined (see Table 1). None of the LOD values exceeded 7 µg/L. Among the twelve quantified elements, four—Cd, Co, Mo, and Pb—were not detected in any of the analyzed samples. The measured results and the percentage of recovery of elements in the reference material are shown in Table 3 to demonstrate the accuracy of the methodology.

Chromium (Cr), copper (Cu), and nickel (Ni) were each detected in less than three individuals (<6% of the sampled population), but only Cu can be quantified with an average concentration of 496 µg/kg ww. Manganese (Mn), mercury (Hg), and selenium (Se) were more frequently detected across the samples; however, the detection rate of Mn in adult

specimens was low, occurring only in 15% of adult samples, and due to the fact that the concentration measured was below the LOQ, quantification was not possible. In contrast, Hg was found in most samples, but it was only possible to quantify two samples that included one adult individual exceeding the current legal threshold, and another with a concentration approaching that limit. Finally, Se was detected in almost all samples, but 15% of the samples were not quantified, since the values were below the LOQ.

Arsenic (As) and zinc (Zn) were detected in all specimens. Zn concentrations exhibited relatively low variability across the sample set. In this study, it was possible to detect As in all samples, ranging between 0.35–2.54 mg/kg ww.

3.3. Nutritional Assessment of Essential Elements in Meagre Flesh

Seafood is very rich in macro-elements essential to humans. Therefore, the meagre flesh was evaluated, considering the daily recommended intake (DRI) of these elements for an adult male (between 31–50 years old [35]) (Table 4). More detailed information is supplied in Supplementary Table S3.

Table 4. Nutritional values of essential elements in milligrams per 100 g of wet weight of wild meagre flesh in all ontogenic stages (range, means, and standard deviation—SD).

Element	DRI (mg/Day) [35]	100 g of Wet Weight of Wild Meagre Flesh				DRI (%)
		Range (Min.–Max.)	Small (Mean ± SD)	Juvenile (Mean ± SD)	Adult (Mean ± SD)	
Ca	1000	5.582–19.591	7.357 ± 0.715	7.566 ± 2.754	6.945 ± 1.146	0.56–1.96
Fe	8	0.065–1.150	0.154 ± 0.092	0.227 ± 0.224	0.157 ± 0.066	0.81–14.38
K	3400	243.306–381.710	345.555 ± 18.324	346.495 ± 19.454	332.054 ± 27.407	7.16–11.22
Mg	420	20.378–28.241	25.657 ± 1.069	25.473 ± 1.277	23.731 ± 1.273	4.85–6.72
Na	1500	21.096–110.809	37.179 ± 8.934	40.080 ± 11.548	56.903 ± 22.503	1.41–7.39
P	700	137.349–182.891	163.078 ± 8.581	163.942 ± 8.095	151.227 ± 10.517	19.62–26.13
Se *	0.055	0.036–0.073	0.052 ± 0.009	0.053 ± 0.011	0.056 ± 0.011	65.45–132.73
Zn *	11	0.223–0.865	0.303 ± 0.040	0.298 ± 0.047	0.328 ± 0.135	2.03–7.86

Daily Reference Intake (DRI), * Elements quantified.

The concentrations used for the calculations were obtained through semi-quantification, except for Se and Zn. Thus, the presented values are very close to real values. For the majority of the elements, the intake is below 10% of DRI per 100 g of *ww* of wild meagre flesh (see Table 4); on the other hand, the quantity of Se present can provide at least half of the total DRI for an adult male.

3.4. THQ, HI, and TR for Each Class of Individuals

For all individuals where the concentration of elements is above the LOD for quantification, Target Hazard Quotients (THQs) were calculated to assess potential non-carcinogenic risks associated with human consumption. To calculate THQ for As (inorganic) and Hg (methyl), it was considered that 5% of total As is inorganic [36] and 95% of Hg is methyl-Hg, because usually, almost all Hg found in aquatic predators is methyl-Hg; these percentages allowed us to estimate THQ values in this study. The THQ values calculated for As were particularly elevated across all ontogenic stages (Table 5), presenting high carcinogenic potential in more than 70% of samples ($THQ_{As} > 1$). Only two Hg samples were quantified (above the LOQ), and only one (fewer than 2% of the samples) is above the safety limit; the THQ for Hg was only calculated for these two samples, and the values were well above 1 (9.86 and 13.27). In contrast, the THQs for Se and Zn were substantially lower, suggesting minimal concern for human health (see Table 5). The Hazard Index (HI), representing the cumulative risk from multiple elements, was also high for all meagre size classes, with As and Hg contributing most significantly to the overall value; only 9% of the samples had a

value below 1, representing a low risk of carcinogenic potential. More detailed information can be seen in Supplementary Table S4.

Table 5. Target Hazard Quotient (THQ) for As, Hg, Se, and Zn, calculated using reference doses (RfD) [32] and potential critical effects of high levels of each element, the Hazard index (HI), and the Target Carcinogenic Risk (TR) for As, calculated using the cancer slope factor (CSF) [37] for each ontogenic stage, calculated using the concentration of each element in the muscle of wild meagre (range, means, and standard deviation).

Element	Range (Min.–Max.)	Small	Juvenile	Adult	RfD (mg/kg Day) [32]	Critical Effect Systems
THQ						
As (inorganic) *	0.616–4.495	1.971 ± 0.707	1.585 ± 0.907	1.446 ± 0.600	6×10^{-5}	Cardiovascular, Endocrine
Hg (methyl-) *	9.859–13.274	-	-	11.567 ± 1.707	1×10^{-4}	Developmental, Nervous
Se	0.152–0.309	0.222 ± 0.038	0.224 ± 0.045	0.237 ± 0.046	5×10^{-3}	Dermal, Hematologic, nervous
Zn	0.016–0.061	0.022 ± 0.003	0.021 ± 0.003	0.023 ± 0.010	3×10^{-1}	Hematologic, Immune
HI	0.0647–14.298	2.214 ± 0.725	1.808 ± 0.928	2.803 ± 3.572	-	-
TR					Cancer slope factor (CSF) oral (mg/kg day) [37]	
As	(0.555–4.045) $\times 10^{-4}$	(1.773 ± 0.636) $\times 10^{-4}$	(1.426 ± 0.817) $\times 10^{-4}$	(1.301 ± 0.540) $\times 10^{-4}$	1.5	

* The concentration of As used corresponds to 5% of the total As according to [36]. The concentration of Hg used corresponds to 95%.

An additional factor of critical importance is the carcinogenic risk, especially considering that meagre is frequently captured and consumed by humans. Using the oral cancer slope factor (CSF), the carcinogenic risk was estimated for As (Table 5). The result indicates that the carcinogenic risks associated with As is relevant, exceeding the generally accepted safety threshold of 1×10^{-4} in more than 60% of the samples, thus highlighting a potential health concern linked to chronic exposure.

4. Discussion

The semi-quantification of essential elements in the meagre muscle allowed us to unveil their approximate concentrations. Several elements were detected in high concentrations, such as K, Na, and P, which was expected because they are macro-elements present in fish matrices. Therefore, meagre follows the same pattern as other fish species, being a good source of these macronutrients in the human diet. Other non-essential elements, such as Ti, B, and Al, were detected in all or almost all samples in this study. The detection of Sr and V was interesting, but the low prevalence in this study did not allow us to draw any detailed conclusions. Quantification was performed for toxic elements and elements that can be toxic, depending on their respective concentrations. According to current European Commission regulations [38], the maximum permissible concentrations in fish tissue are defined for only four of these elements: Cd (0.05 mg/kg, ww), Pb (0.30 mg/kg, ww), Hg (0.50 mg/kg, ww), and inorganic As (0.1 mg/kg ww). As the calculated LOQs for these elements fall well below the regulatory thresholds, the absence of Cd and Pb in all samples suggests that the consumption of meagre from the study area does not pose a health risk with regard to these two contaminants.

In some samples, it was possible to detect Ni according to our method, but it was not quantified since the concentration was below the LOQ. This result is also important since it reveals the low concentration found for this element, taking into consideration the fact that high concentrations can be harmful to humans. Hg was detected in all samples in this study. Although only present in two samples, the values fell above the LOQ and can therefore be quantified. It was found that one sample revealed values above the safety limit (0.5 mg/kg ww), and another revealed a value very close to the safety limit. As, Se, and Zn were quantified. Considering the values for As, which fell within the range of 0.35–2.54 mg/kg ww, this result should be evaluated with caution. The first indication is that As should be prevalent in the waters of the Tagus estuary. The European Union, in September 2025, defined a regulatory limit for seafood products by amending Regulation (EU) 2023/915 [38] to include limits for inorganic As in seafood. The new regulatory limit proposed a maximum allowable concentration of 0.1 mg/kg ww for all species not listed under Section 3.4.5.2 of the above-mentioned regulation (European Union, in September 2025 [38]). The concentration of As measured in this study refers to the total As; however, from other studies, it is established that inorganic As is approximately 3–5% of the total As [36]. Therefore, when this ratio is applied, and considering the highest percentage found in the literature (5%), our results revealed that values ranged between 0.010–0.127 mg/kg ww. Only one individual was above this limit (0.1 mg/kg ww). Nevertheless, several samples presented concentrations of predicted inorganic As above 0.08 mg/kg ww.

Chiesa et al. [26] and Maulvault et al. [27] found levels of As and other potentially harmful metals in mussels, clams, and mullets (*Liza aurata*) from the Tagus estuary, which are aligned with the results obtained in this work.

To the best of our knowledge, no previous studies have examined elemental concentrations in meagre in relation to gender or ontogenic stage. Chaguri et al. [28] analyzed wild specimens across two ontogenic stages: smaller individuals (median weight ~0.6 kg), which are below the size range included in this study, and juveniles (2–7 kg), which are partially comparable to the juvenile group analyzed herein. However, due to methodological discrepancies, particularly the expression of concentrations on a dw basis in their study versus ww in the current work, a direct comparison with existing regulatory limits, which are defined on a ww basis, is not possible.

The results obtained for total As and Hg indicate "safety" for consumers, but it should be noted that both toxic elements were detected in all samples in this study, regardless of ontogenetic stage or even gender. In order to calculate the THQ, and in the cases of As and Hg, only the concentration values of inorganic As and methyl-Hg were used. In this work, we emphasized that the total concentrations of these two elements were calculated according to the literature [33]. Thus, a value of 5% of total As was considered as inorganic As and 95% of total Hg as methyl-Hg in order to estimate THQ values. These results are thus an average prediction because the concentrations of inorganic As and methyl-Hg were not specifically determined.

Considering THQ values for each quantified element, our results revealed high values for $THQ_{inorganicAs}$ and $THQ_{methyl\text{-}Hg}$ in at least 70% of the samples regarding As, and for two samples for Hg, which indicate potential health hazards. For Se and Zn, our results presented low THQ values, indicating residual potential health risks associated with these elements.

When $THQ_{inorganicAs}$ and $THQ_{methyl\text{-}Hg}$ are both above one, the HI value is above one and will immediately mark the samples as being potential health hazards. When the sum of $THQ_{inorganicAs}$ and $THQ_{methyl\text{-}Hg}$ is above one, it will also mark the samples as potential health hazards. Considering this, 90% of all samples in this study present an HI value

above 1. These facts support the importance of frequent monitoring in order to mitigate potential health risks for consumers.

TR values for As showed a high target cancer risk for at least 60% of all samples in this study.

Analyzing the results obtained by comparing the concentrations measured with regulatory limits, as well as with THQ, HI, and TR values, it is important to highlight that the results disclosed here do not allow us to predict and assess the imminent dangers associated with the consumption of meagre from this location, even in small quantities.

The nutritional value of the macro-elements in the edible portion of this species showed that it is a good source of these elements, as almost all of them represent approximately 10% of the RDI. The exception is Se, since consumption of 100 g ww of an edible portion of meagre allows consumers to ingest half of the total DRI for this element.

5. Conclusions

In this study, a total of 34 elements were analyzed in the muscle tissue of 55 meagre specimens captured in the Tagus estuary. Elemental analysis was conducted using microwave-assisted digestion followed by ICP-OES. The results revealed that less than 2% of individuals presented a Hg concentration exceeding the current regulatory limit of 0.5 mg/kg ww. Furthermore, using the new proposed regulatory limit for inorganic As in fish samples, one specimen in this work would surpass that threshold (less than 2% of the total individuals).

The values for essential elements, such as Fe, K, Mg, Zn, and Se, show that the meagre is a species that can be considered an important food resource for a balanced diet, being rich in essential elements, with approximately 10% of the DRI in 100 g ww of meagre flesh, except for Se, which can meet the total DRI in 100 g ww.

The results of the obtained THQ, HI, and TR, as well as the As and Hg concentrations measured, suggest that it is wise to prioritize monitoring these elements, especially in seafood associated with estuarine habitats such as the Tagus, to ensure consumer safety.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/oceans7010003/s1, Table S1. Results ICP-OES analysis in mg/kg wet weight for semi-quantification of each element in all samples of meagre muscle; Table S2. Results ICP-OES analysis in mg/kg ww for quantification of each element in all samples of meagre muscle; Table S3. Nutritional values of essential elements in milligrams per 100 g of wet weight of wild meagre flesh in all meagre samples; Table S4. Target Hazard Quotient (THQ) for As, Hg, Se and Zn, Hazard index (HI) and Target carcinogenic Risk (TR) for As, calculated for each sample using concentration of each element in muscle of wild meagre.

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